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Pharmacy students' perception and knowledge of chat-based artificial intelligence tools at a Nigerian University

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Abstract

Background Chat-based Artificial Intelligence (AI) tools, such as ChatGPT®, are becoming integral to various aspects of pharmacy education. However, their integration into the curriculum faces challenges due to students' varying levels of knowledge and perceptions. This study aimed to evaluate pharmacy students' knowledge and perception of chat-based AI tools at Afe Babalola University, Ado-Ekiti, Nigeria (ABUAD). It also assessed their familiarity with these tools and their usage patterns.

Method A cross-sectional online survey was conducted from March to April 2024 among undergraduate pharmacy students, selected through random sampling. Student knowledge was categorised as good or poor while perception was grouped into positive or negative. Data analysis was conducted using Statistical Product and Service Solutions version 27.

Results A total of 252 students participated in this study with the majority being female (72.2%). Most students (88%, $n = 222$) were familiar with chat-based AI tools, with ChatGPT® being the most commonly used (82.8%) for assignments and studying. Students generally showed a positive perception of the tools, with 85.3% believing it enhances academic performance. Concerns were raised about potential distractions (65.7%) and the risk of academic dishonesty (65.1%). Students with prior AI education ($p < 0.001$), higher levels of study ($p = 0.011$), and prior awareness ($p < 0.001$) demonstrated significantly higher knowledge scores.

Conclusion Pharmacy students at ABUAD demonstrated good knowledge of chat-based AI tools and generally positive perceptions towards its use. The study underscores the need to integrate AI education into the pharmacy curriculum to address knowledge gaps and better prepare students for future technological advancements.

Keywords Artificial intelligence, Pharmacy education, Student perception, Chat-based AI tools, Nigeria

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Introduction

Artificial Intelligence (AI) is transforming various sectors, including healthcare, finance, and education, with intelligent voice assistants like Siri, and Alexa as well as chat-based AI tools like ChatGPT® becoming integral to modern communication and problem-solving. These tools, which simulate human cognition, are redefining how tasks are performed and decisions are made in real time [1]. The ability of AI to offer personalized support and interactive learning makes it a valuable asset in education, including pharmacy training [2].

As AI continues to evolve, its integration into pharmacy education marks a new phase of technological advancement, reshaping how future pharmacists learn and prepare for their careers [3]. Chat-based AI tools, in particular, have shown immense potential in enhancing learning by providing real-time responses, supporting data-driven decisions, and personalizing educational content.

Adaptive learning systems, powered by AI, personalize content based on individual learning styles, strengths, and weaknesses [4]. This tailored approach deepens students' understanding and retention of complex pharmaceutical concepts, fostering more effective learning. Additionally, AI-driven simulation and virtual reality technologies offer immersive learning environments where pharmacy students can practice clinical skills in realistic settings, enhancing their practical proficiency and confidence [5].

In an era dominated by big data and precision medicine, the ability to make data-driven decisions is critical for pharmacists. AI algorithms can analyze large datasets, uncovering insights that might be overlooked by human analysis. Integrating AI into pharmacy education equips students with these analytical tools, enabling more informed decision-making in patient care, drug interactions, and treatment outcomes [6]. This capability is increasingly important as healthcare becomes more data-driven and personalized.

Beyond personalized learning and data analysis, AI also streamlines administrative tasks in education. AI-powered platforms and chatbots, for instance, automate grading and feedback processes, allowing educators to focus more on personalized student interactions [7]. The primary goal of pharmacy education is to improve patient care, and AI tools empower future pharmacists to deliver precise, efficient, and personalized healthcare. For example, AI facilitates the calculation of optimal drug dosages tailored to individual patient profiles based on genetic and lifestyle factors, improving patient outcomes and advancing personalized medicine [6].

In clinical settings, AI aids healthcare providers by analyzing patient data to diagnose diseases and recommend treatments [8]. Platforms like IBM Watson Health, driven

by AI, offer personalized treatment recommendations based on current clinical guidelines and patient-specific data, enhancing decision-making [9]. Moreover, the role of AI in personalized medicine is significant as it can tailor treatments to individuals' genetic profiles and health data, optimizing therapeutic outcomes and minimizing adverse effects.

Chat-based AI tools, such as ChatGPT® and other conversational AI systems, are emerging as valuable resources in pharmacy education. These tools simulate human-like interactions and provide real-time responses, offering immediate support for students in various aspects of their learning.

The use of chat-based tools in pharmacy education is particularly effective in fostering active learning. Students can ask questions and receive tailored, detailed explanations of complex pharmaceutical concepts. Chat-based tools help students clarify doubts, solve problems, and conduct research, making them valuable for both individual and group study sessions. For instance, pharmacy students often use tools like ChatGPT® for assignments, studying, and accessing vast databases of medical knowledge [5].

Moreover, these tools offer constant availability, providing round-the-clock support. This ensures that students can continue learning outside traditional classroom hours, fostering continuous engagement. Chat-based AI tools also help bridge gaps in knowledge by offering personalized learning pathways based on students' queries and responses [6].

While chat-based AI tools significantly enhance learning, there are concerns related to their potential for distraction and misuse, particularly regarding academic dishonesty. The accessibility of instant answers can encourage shortcuts rather than deep learning, which raises questions about how these tools should be regulated within educational settings.

Despite the potential benefits, the integration of AI, including chat-based tools, into pharmacy curricula varies widely across institutions. Many schools lack standardized approaches for incorporating AI into their programs, leading to unequal exposure for students [10]. Developing AI-related skills is critical for pharmacy students, yet challenges such as limited technological infrastructure, faculty expertise, and resources hinder widespread adoption [11].

Understanding how pharmacy students perceive and use AI tools, including chat-based systems, is essential for effective integration. Surveys conducted in Nigeria and other regions reveal that students generally hold positive attitudes towards AI, recognizing its potential to enhance their learning [12]. However, concerns about privacy, algorithm complexity, and the need for additional training remain prevalent [13–15]. These concerns highlight

the importance of developing educational strategies that address both the benefits and potential drawbacks of AI tools.

Despite the growing interest in the role of AI in education, there are notable gaps in the literature. Much of the existing research focuses on the theoretical benefits of AI rather than its practical application, particularly in pharmacy programs. There is limited empirical data on how AI tools, especially chat-based systems, are currently being used in pharmacy education, especially in developing countries like Nigeria.

Additionally, research on pharmacy students' perceptions and perception toward chat-based AI tools remains scarce. While studies in developed countries show positive attitudes toward AI [12, 14], little is known about how pharmacy students in developing countries view and interact with these technologies. Understanding these perceptions is vital for developing strategies to effectively incorporate AI tools into educational programs.

This study aimed to evaluate pharmacy students' knowledge and perception of chat-based AI tools at Afe Babalola University, Ado-Ekiti, Nigeria (ABUAD). It also assessed their familiarity with these tools and their usage patterns.

Methods

Study design and setting

This study was a cross-sectional observational study. The study was conducted online through Google Forms at Afe Babalola University, Ado-Ekiti (ABUAD), located in Ekiti State, southwestern Nigeria. Ado-Ekiti is the capital of Ekiti State. ABUAD, a leading private university, has a student population of over 8,000 [16] and is renowned for its state-of-the-art teaching facilities, including advanced e-learning platforms and electronic boards.

The College of Pharmacy at Afe Babalola University began as a faculty within the College of Medicine and Health Sciences in August 2019. It offers a six-year Doctor of Pharmacy (Pharm.D.) program and is committed to academic excellence in pharmaceutical sciences. The college's vision is to promote good health and wellness through innovative education and impactful research, positioning itself as a leader in the field [17]. The data collection period spanned from March to April 2024.

Participants and sample size

The study targeted undergraduate pharmacy students at ABUAD. Participants were randomly selected from the student population of the College of Pharmacy. Inclusion criteria were current enrollment as a pharmacy student and consent to participate in the study. Students who did not consent were excluded. The sample size was determined using the Raosoft® online calculator. With a total

population of 730 pharmacy students and a 95% confidence level, the required sample size.

252 students. This sample size was deemed sufficient to ensure statistical reliability and representativeness.

Study tool and item scoring

A 26-item online questionnaire was used to collect data on the students. The questionnaire had 3 sections. The first section assessed the sociodemographic parameters using 6 items including age, study level, and ethnicity. The second section assessed students' knowledge, usage, and purpose of usage of AI applications in pharmacy education. The section was assessed using 9 questions made up of multiple choice as well as yes and no questions. The third section assessed the students' perception of chat-based AI tools. To assess students' perceptions of chat-based AI tools in pharmacy education using mean perception scores, a methodological approach was adopted. This section comprised of 11 questions, each employing a 5-point Likert scale (ranging from Strongly Disagree to Strongly Agree).

Knowledge scores were calculated based on correct answers to the questions in the second section of the questionnaire. Each correct answer was awarded 1 point, and incorrect answers received 0 points. Based on their total scores, students were classified as having either poor knowledge (<50%) or good knowledge ($\geq 50\%$).

Perception scores were calculated by summing the responses to 11 Likert-scale items. Each response ranged from 1 (Strongly Disagree) to 5 (Strongly Agree). An overall perception score for each student was computed by summing these values. A score $\geq 50\%$ was classified as a positive perception, while scores <50% were considered negative.

Data collection

Students were randomly selected for the study by assigning numbers to all students in the college and using a random number generator to choose the required participants. The online questionnaire link was shared with selected students in each WhatsApp group. Reminders were sent to the target population every week throughout the study period. Participation was entirely voluntary, and data collection remained anonymous. Respondents were informed about the survey's purpose and nature, provided informed consent, and were given the option to withdraw at any point.

Data analysis

Descriptive statistics (frequencies, percentages, means, and standard deviations) were used to summarize demographic characteristics and students' responses regarding their knowledge and perceptions of AI tools.

Table 1 Socio-demographics of the study participants

A. Demographics		Frequency (N=252)	Percent
Age (Mean ± SD)		19.17 ± 1.813 years (Range: 16–27 years)	
Religion	Christianity	219	86.9
	Islam	32	12.7
	Others	1	0.4
Gender	Male	70	27.8
	Female	182	72.2
Ethnicity	Yoruba	104	41.3
	Igbo	82	32.5
	Hausa	5	2.0
	Others	61	24.2
Religion	Islam	32	12.7
	Christianity	219	86.9
	Others	1	0.4
Level of study	100–300 Level	144	57.1
	400–600 Level	108	42.9
Have you received any prior education or training on AI in pharmacy Education (N=252)		Yes	41
		No	211
			16.3
			83.7

Key: AI: Artificial Intelligence, SD: standard deviation

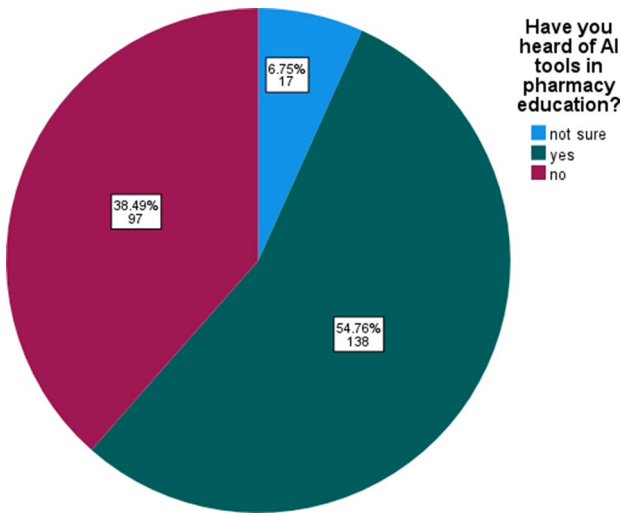


Fig. 1 Percentage of students that have heard of AI tools in pharmacy education

Inferential statistics included the use of independent sample t-tests and one-way ANOVA to compare AI knowledge scores across demographic subgroups (age, gender, level of study, and prior AI training). Kolmogorov-Smirnov test was performed to assess data normality ($p=0.08$), which justified the use of parametric tests. Subgroup analyses were conducted to compare AI knowledge and perception scores between different levels of study and AI training. The independent sample t-test was used for two-group comparisons, and one-way ANOVA was applied to compare more than two groups.

Results
Socio-demographics

The respondents' ages ranged from 16 to 27 years, with an average age of 19.17 ± 1.813 years. The majority identified as Christians (86.9%, $n=219$), and the gender distribution showed that 70 respondents (27.8%) were male. Regarding ethnicity, 41.3% ($n=104$) were Yoruba, while 32.5% ($n=82$) were Igbo. The respondents' levels of study were divided into two categories: 57.1% ($n=144$) were in 100–300 level, and 42.9% ($n=108$) were in 400–600 level. Regarding prior education or training on AI in pharmacy education, only 16.3% ($n=41$) had received training, while 83.7% ($n=211$) had not. A summary of the socio-demographics is provided in Table 1.

AI tools and purpose of use among university students

A majority of students (54.8%) reported being aware of AI tools in pharmacy education, while 6.8% were unsure (Fig. 1). Over 88% ($n=223$) stated that they had used an AI tool before (Fig. 2). The most commonly used tool was ChatGPT® (82.8%, $n=202$), followed by Grammarly® (41.4%, $n=101$) and Siri® (29.9%, $n=73$). Other AI tools, such as QuillBot® (27.5%, $n=67$) and Snapchat AI® (4.1%, $n=10$), were also reported. Less commonly used AI tools included Admissions IVY ChatBot®, InVideo®, Superintelligence®, and several others, each used by a single respondent. In terms of purpose, 80.0% ($n=195$) of respondents used AI tools for completing assignments, and 79.6% ($n=194$) for studying. Other uses included obtaining drug information (51.6%, $n=126$), writing research articles (43.1%, $n=105$), and personalized learning recommendations (41.0%, $n=100$). Fewer students used AI

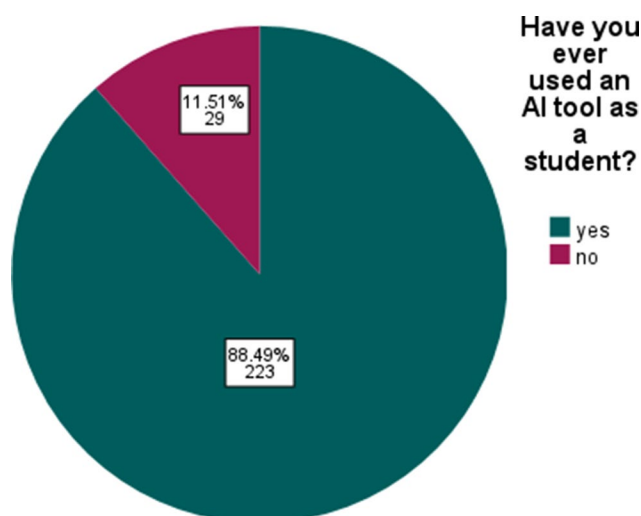


Fig. 2 Percentage of students who have used an AI tool

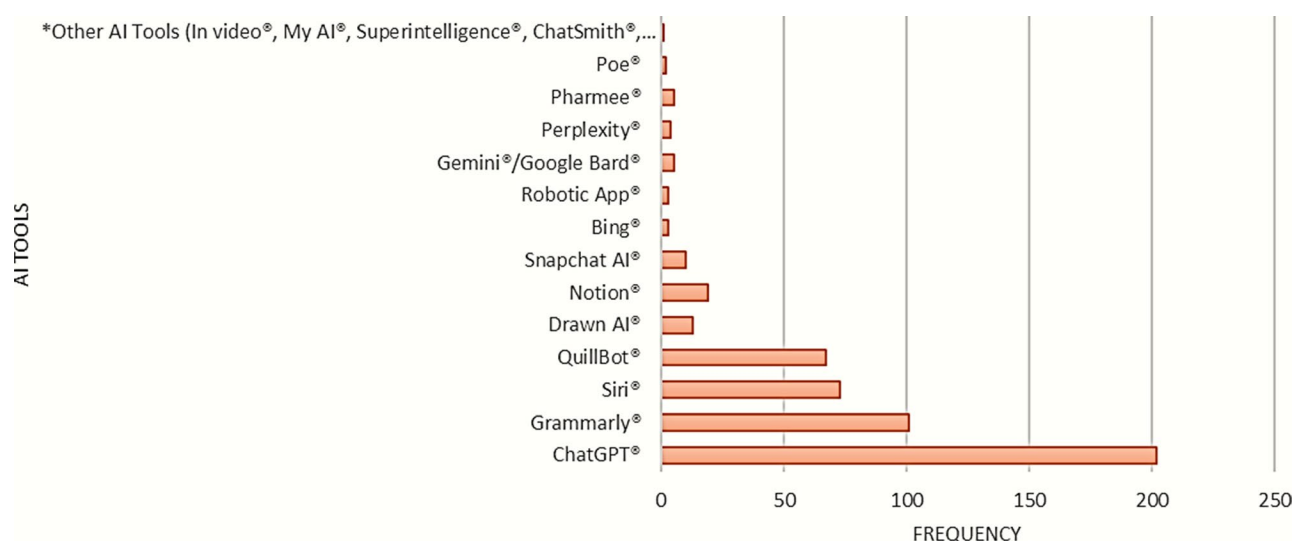


Fig. 3 AI tools used among the university students

tools for intelligent content generation (19.7%, $n=48$), collaborative learning platforms (13.5%, $n=33$), or virtual labs (7.0%, $n=17$). The details are shown in Figs. 3 and 4.

Perception of students towards chat-based AI tools in pharmacy education

The survey revealed mixed perceptions among students regarding the use of chat-based AI tools in pharmacy education. Concerns about engagement and distraction were prevalent, with 65.7% worried that AI tools might reduce their engagement during lectures, and 48.2% noting that AI use in online lectures could lead to distraction from home-related activities.

Despite these concerns, a majority of students (58.3%) believed that chat-based AI tools are essential for the future of pharmacy education, and 85.3% perceived that these tools positively impact their academic performance.

Self-discipline was considered crucial, with 73.9% stating that using AI tools during class requires it, while 65.1% felt that AI tools could increase the likelihood of academic dishonesty.

Opinions on class structure were divided, with 47.6% indicating that classes became less structured with AI tool integration. However, 88.1% recognized the potential of AI tools to enhance the learning experience, and 88.5% appreciated the increased accessibility to educational resources provided by these tools. A portion of the respondents (38.3%) expressed concerns about inefficiencies in coursework due to AI tools, yet 64.2% believed all pharmacy students should receive education in AI. Overall, 88.5% had a positive perception of chat-based AI tools in pharmacy education. The findings are summarized in Table 2.

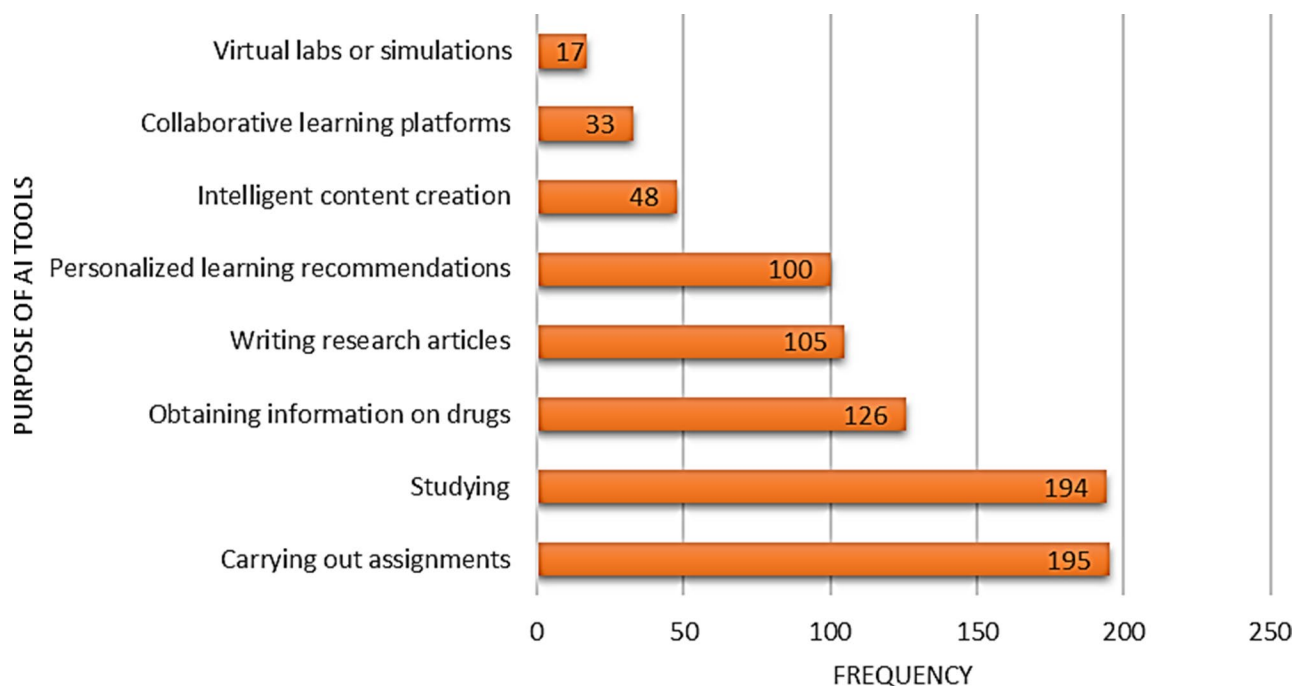


Fig. 4 Purpose of AI tools used among the university students

Table 2 Perception of students towards chat-based AI tools in pharmacy education

S/N	Statements	Responses (%)				
		SA	A	N	D	SD
1.	AI tools use reduces engagement/concentration from students during lectures	23 (9.1)	83 (32.9)	65 (25.8)	53 (21.0)	28 (11.1)
2.	AI is the future of pharmacy education	55 (21.8)	92 (36.5)	67 (26.6)	21 (8.3)	17 (6.7)
3.	I believe AI tools contribute positively to my academic performance in pharmacy education	112 (44.4)	103 (40.9)	32 (12.7)	5 (2.0)	-
4.	Use of AI tools in giving online lectures makes it easier to get distracted by things going on in the home	30 (11.9)	84 (33.3)	78 (31.0)	43 (17.1)	17 (6.7)
5.	Using AI tools during class requires self-discipline	78 (31.0)	108 (42.9)	45 (17.9)	11 (4.4)	10 (4.0)
6.	Students are more likely to engage in academic dishonesty when using AI tools in taking tests or examinations	64 (25.4)	100 (39.7)	53 (21.0)	16 (6.3)	19 (7.5)
7.	Classes are less structured when instructors use AI tools to deliver lectures	27 (10.7)	71 (28.2)	80 (31.7)	44 (17.5)	30 (11.9)
8.	AI tools can enhance the learning experience of students in pharmacy education	107 (42.5)	115 (45.6)	24 (9.5)	3 (1.2)	3 (1.2)
9.	AI tools use in learning can bring increased accessibility to educational resources	113 (44.8)	110 (43.7)	24 (9.5)	1 (0.4)	4 (1.6)
10.	AI tools use in learning can bring inefficiency in course work	23 (9.1)	51 (20.2)	78 (31.0)	55 (21.8)	45 (17.9)
11.	All pharmacy students should receive teaching in AI during their study	70 (27.8)	94 (37.3)	63 (25.0)	15 (6.0)	10 (4.0)
	Positive Perception ($\geq 50\%$ of mean perception score)	223 (88.5)				
	Negative Perception ($< 50\%$ of mean perception score)	29 (11.5)				

Key: AI=Artificial Intelligence; SA=Strong Agree-5; A=Agree-4; N=Neutral-3; D=Disagree-2; SD=Strongly Disagree-1

Knowledge of students on chat-based AI tools in pharmacy education

The assessment of students' knowledge regarding chat-based artificial intelligence (AI) tools in pharmacy education was also conducted. A notable 64.7% ($n=163$) incorrectly believed that these tools function using a small amount of data pre-processed by humans.

Meanwhile, 37.3% correctly disagreed with the idea that AI integration is aimed at replacing traditional teaching methods.

Regarding the capabilities of AI tools, 82.1% ($n=207$) recognized that these tools create personalized learning experiences based on individual needs and performance metrics. However, only 19.4% ($n=49$) correctly disagreed

with the statement that AI-generated information is always accurate. Overall, 20.6% ($n=52$) of students demonstrated poor knowledge of chat-based AI tools in pharmacy education. This has been detailed in Table 3.

Relationship between student demographics, overall perception, and knowledge of chat-based AI tools in pharmacy education

The relationship between various demographic variables and students' knowledge of AI tools was analysed. Male students (27.8%) had a mean knowledge score of 3.60 ± 1.256 , while female students (72.2%) had a slightly lower mean score of 3.51 ± 1.341 . However, the p -value (0.632) indicated no significant gender-based difference in knowledge scores.

In terms of age, the p -value of 0.244 showed no significant differences in knowledge scores between different age groups, though students aged 20–23 years had slightly higher scores compared to those over 23 years old.

A significant difference was observed based on the level of study, with higher-level students scoring significantly higher ($p=0.011$). Students with prior AI training also had notably higher knowledge scores than those without training, and this difference was statistically significant ($p<0.001$).

Students who were aware of AI tools in pharmacy education and those who had used AI tools had significantly higher knowledge scores than those who were unaware or had not used AI tools ($p<0.0001$). These results are summarized in Table 4.

Discussion

Artificial Intelligence is increasingly influencing various sectors, including education and healthcare. In pharmacy education, AI technologies offer promising opportunities to enhance learning, research, and practice [3]. These technologies can provide personalized learning experiences, improve access to information, and support the development of critical thinking and analytical skills. In Nigerian universities, where pharmacy education is rapidly evolving, understanding students' perceptions and knowledge of AI is crucial. As AI tools become more integrated into educational frameworks, it is essential to assess how students view these technologies and how well they understand their applications and implications. This study explored the perception and knowledge of chat-based AI tools among pharmacy students at Afe Babalola University, Ado-Ekiti. The findings of this study reveal significant insights into pharmacy students' knowledge and perceptions of AI in pharmacy education.

Our study revealed a high level of engagement with chat-based AI tools among pharmacy students at Afe Babalola University, with over 88% of students reporting prior use of AI tools. This finding is consistent with research conducted by Mosleh et al. (2023) among medical and pharmacy students in Jordan, where 81% had heard of AI programs, and 42.1% had used them [18]. Similarly, a multinational study conducted in Jordan, Egypt, Saudi Arabia, and Palestine [14] reported high levels of AI awareness and usage among pharmacy students, indicating a broader trend of increasing AI adoption in healthcare education globally. The high engagement observed in our study suggests that pharmacy students

Table 3 Knowledge of students on chat-based AI tools in pharmacy education

S/N	Statement			Correct (%)	Incorrect (%)
1	AI tools work by using small amount of data already processed by a human	Yes (%)	163 (64.7)	163 (64.7)	89 (35.3)
		No (%)	22 (8.7)		
		I don't know (%)	67 (26.6)		
2	Integration of chat-based AI tools into pharmacy education is aimed primarily at replacing traditional teaching methods	Yes (%)	111 (44.0)	141 (56.0)	111 (44.0)
		No (%)	94 (37.3)		
		I don't know (%)	47 (18.7)		
3	AI tools can be employed to create personalized learning plans for pharmacy students based on their individual needs and performance	Yes (%)	207 (82.1)	207 (82.1)	45 (17.9)
		No (%)	9 (3.6)		
		I don't know (%)	36 (14.3)		
4	AI-generated data on pharmacy-related information are always accurate	Yes (%)	49 (19.4)	203 (80.6)	49 (19.4)
		No (%)	120 (47.6)		
		I don't know (%)	83 (32.9)		
	Remark			Score	Frequency (%)
	Poor knowledge			< 50%	52 (20.6)
	Good knowledge				200 (79.4)

Key: AI= Artificial Intelligence

Table 4 Relationship between student demographics, overall perception and knowledge of chat-based AI tools in pharmacy education

Variables		Frequency (%)	Mean score $\pm$ SD	p-value
Gender	Male	70 (27.8)	3.60 $\pm$ 1.256	0.632 ^a
	Female	182 (72.2)	3.51 $\pm$ 1.341	
Age	16–19 years	139 (55.2)	3.41 $\pm$ 1.318	0.244 ^b
	20–23 years	110 (43.7)	3.69 $\pm$ 1.305	
	Above 23 years	3 (1.2)	3.67 $\pm$ 1.528	
Level of study	100–300	144 (70.8)	3.35 $\pm$ 1.314	0.011^{a*}
	400–600	108 (29.2)	3.78 $\pm$ 1.285	
Have you received any prior education or training on AI in pharmacy Education?	Yes	41 (16.3)	4.27 $\pm$ 1.141	< 0.001^{a*}
	No	211 (83.7)	3.39 $\pm$ 1.303	
Have you heard of AI tools in pharmacy education?	Yes	138	4.23 $\pm$ 0.969	< 0.001^{a*}
	No	97	2.72 $\pm$ 1.170	
Have you ever used an AI tool as a student?	Yes	223	3.71 $\pm$ 1.204	< 0.001^{a*}
	No	29	2.17 $\pm$ 1.365	
Negative Perception		29 (11.5)	3.38 $\pm$ 1.474	0.497 ^a
Positive Perception		223 (88.5)	3.56 $\pm$ 1.296	

Key: SD: Standard deviation; ^a: Independent Sample T-test; ^b: One Way Analysis of Variance; *: Statistically significant value, $p \leq 0.05$

are becoming more accustomed to integrating AI into their academic activities, aligning with global trends.

The study also explored how various chat-based AI tools are utilized by pharmacy students. ChatGPT® was the most frequently used tool, with 80.2% of students reporting its use. This is in contrast to the study by Oluwadiya et al. (2023) in 10 universities in Nigeria where Grammar Checkers® was the most used AI tool followed by ChatGPT® [13]. The preference for ChatGPT® in our study may reflect its versatility and widespread recognition as a conversational AI tool capable of handling diverse tasks, from academic writing to information retrieval. This highlights the evolving landscape of AI tool adoption, where different regions or institutions might favour specific tools based on their functionality or ease of use.

A significant portion of our students reported using AI tools for academic purposes, such as assignments (77.4%) and studying (77.0%). This is a notable increase compared to the findings of Mosleh et al. (2023), where only 44.2% of students used AI programs for studying and 38.9% for assignments [18]. The higher utilization rates in our study suggest that AI tools are becoming integral to the educational activities of pharmacy students, potentially enhancing their learning efficiency and academic performance. However, this widespread usage also raises concerns about over-reliance on chat-based AI tools, which may hinder the development of critical thinking and problem-solving skills, as students might be tempted to delegate complex tasks to AI without fully engaging with the material themselves.

Moreover, our study found that 50.0% of students used chat-based AI tools to obtain drug information, and 41.7% employed them for writing research articles. This

reflects the growing impact of AI on research and information retrieval in pharmacy education, aligning with Tai's assertion that AI can significantly enhance the quality and accessibility of drug information [19]. However, this reliance on AI for critical information retrieval tasks necessitates robust training on evaluating AI-generated content to prevent misinformation and ensure the accuracy of the data used in clinical decision-making.

Despite the benefits of AI, our study identified several misconceptions among students.

A significant misconception was that the integration of AI tools into pharmacy education is primarily aimed at replacing traditional teaching methods. This belief could foster resistance to AI adoption, as students may fear that AI will supplant human educators. Our findings are consistent with those reported by Sit and colleagues, who observed similar concerns among medical students about AI potentially replacing teachers and diminishing the role of human interaction in education [20]. In contrast, Almasri. (2024) emphasized that the role of AI in education should be seen as complementary, enhancing traditional teaching methods by offering personalized and adaptive learning experiences [21]. The persistence of this misconception underscores the need for clear communication about the supportive role of AI in education, rather than as a replacement for human educators.

Our study also found that some students believed that AI-generated data on pharmacy-related information is always accurate. This misconception is particularly concerning given the potential risks associated with relying on inaccurate or biased AI-generated information. This finding mirrors those of Zhai et al. (2024) [22], who reported that students often overestimate the reliability of AI-generated data, failing to recognize the possibility

of errors or biases inherent in AI systems. Similarly, Nguyen. (2023) highlighted the importance of educating students on the limitations of AI and the need for critical evaluation of AI-generated content to prevent the dissemination of misinformation in healthcare settings [23].

A notable proportion of students in our study also believed that the use of chat-based AI tools during lectures could reduce student engagement and concentration. This may arise from experiences where AI tools were not effectively integrated into the learning process, leading to distractions rather than enhancing focus. Taherian and Kim, (2022) [24] observed similar concerns among students in their study, where there was a reported negative effect in improper use of AI tools in student teaching leading to decreased engagement. However, a study by Miao and Ma, (2022) suggested that when AI tools are thoughtfully integrated, they can actually increase engagement by providing interactive and personalized learning experiences [25]. This highlights the need for proper training and guidelines on the effective use of AI tools in educational settings.

The persistence of these misconceptions has significant implications for pharmacy education. If left unaddressed, they could hinder the effective integration of AI into educational practices and limit students' ability to fully benefit from these technologies. Misconceptions about the role of AI in education and its capabilities could lead to over-reliance on these tools, resulting in superficial learning and a lack of critical engagement with the material.

To counter these misconceptions, pharmacy education programs should incorporate comprehensive AI literacy modules that address the capabilities, limitations, and ethical considerations of AI in healthcare. These modules should emphasize the importance of human oversight in the use of AI tools and promote critical thinking skills that enable students to evaluate AI-generated content effectively. Additionally, educators should foster an understanding of AI as a complementary tool that enhances, rather than replaces, traditional educational methods and human roles in healthcare.

By addressing these misconceptions through targeted educational interventions, pharmacy schools can better prepare students to navigate the evolving landscape of healthcare technology, ultimately leading to improved learning outcomes and professional readiness.

The majority of the students exhibited good knowledge of chat-based AI tools in pharmacy education dissimilar to findings in a multinational study carried out in African countries where the pharmacy students showed moderate knowledge about AI [14]. Despite the overall good level of knowledge demonstrated by most students, the presence of a substantial minority with poor knowledge highlights an area for improvement. Enhancing AI

literacy through targeted educational programs could bridge these gaps.

The results indicated varying levels of knowledge across different demographic factors. A significant variation was observed between lower-level (100–300) and higher-level (400–600) students. Higher-level students and students who had received formal AI training demonstrated greater knowledge. This aligns with Hornberger et al. (2022) [26], who found that advanced students often have more opportunities to engage with AI technologies through their coursework and practical experiences, thus gaining deeper insights. Their research emphasizes that structured educational interventions are crucial for enhancing students' understanding of AI, highlighting the positive impact of targeted training on knowledge acquisition.

The majority of students (88.5%) displayed a positive perception of chat-based AI tools in pharmacy education similar to findings by Busch et al. (2024) where 58% of pharmacy students showed positive attitudes towards AI use in pharmacy education, reflecting widespread recognition of its benefits [10]. These findings suggest that integrating AI education into the pharmacy curriculum can enhance student preparedness and engagement with emerging technologies [3].

A majority of students (58.3%) believed that AI is the future of pharmacy education. The majority also agreed that AI tools positively contribute to their academic performance supporting findings in Nigeria by Oluwadiya et al. (2023) [13], which highlight a growing recognition of AI's potential to transform educational practices. This positive outlook suggests a general optimism about the role of AI in the future of the field indicating that AI can enhance learning outcomes when integrated into educational settings.

The perception that chat-based AI tools make it easier to get distracted during online lectures was mixed, with 45.2% agreeing. However, the majority of students (73.9%) agreed that using these tools requires self-discipline. This is consistent with findings by Betto et al. (2023) [27], which suggested that while AI can facilitate learning, it may also introduce distractions if not managed properly. Furthermore, while giving lectures using AI tools allows students to learn at their own pace, it is difficult to manage students' concentration, which could prevent them from receiving valuable information from lectures. This implies the need for personal responsibility in utilizing technology effectively and underscores the importance of self-regulation in maximizing the benefits of AI tools. However, in literature, various methods of combating this using machine learning methods based on human face and posture information have been suggested [27].

Concerns about increased academic dishonesty (65.1% agreement) reflect similar worries highlighted among Nigerian students [15] where students with positive perceptions of ChatGPT® were more prone to use it for dishonest academic purposes. This suggests that while these chat-based AI tools offer many advantages, they also pose challenges, including potential misuse in assessments by students. Addressing these concerns is crucial for ensuring the ethical use of AI. Some measures have been suggested in Literature such as the direct questioning approach among Nigerian students [28] as well as the list experiment among Vietnam undergraduates [29] reflecting a cautious approach to understanding its ethical and practical implications. This hesitancy underscores the need for thoughtful integration strategies that balance innovation with academic integrity.

A substantial majority agreed that chat-based AI tools enhance the learning experience and increase accessibility to educational resources, corroborating research by Wollny et al. (2021) [30] that highlights the potential of chat-based AI tools to enrich educational experiences.

Strengths and limitations of the study

One of the key strengths of this study is its detailed examination of pharmacy students' knowledge and perceptions of chat-based AI tools. By providing a comprehensive understanding of current educational trends, the study sheds light on how students interact with and perceive AI technologies. The inclusion of various demographic variables, such as age, gender, and academic level, enhances the depth of analysis, allowing for a nuanced exploration of factors that influence AI understanding. Additionally, the study emphasizes the importance of prior education and practical experience with AI tools, highlighting the value of targeted educational interventions aimed at improving students' AI literacy.

Despite these strengths, the study has several limitations. First, the research is based on a sample drawn from a single university, which may limit the generalizability of the findings to pharmacy students at other institutions or in different regions. While the sample size was statistically adequate, it may not fully represent the broader population of pharmacy students in Nigeria or beyond. To improve generalizability, future research could benefit from including participants from multiple institutions to capture more diverse settings and perspectives.

Another limitation is the reliance on self-reported data collected through online questionnaires. This method can introduce biases, such as socially desirable responses and recall bias, potentially affecting the accuracy of the information provided. Additionally, the study mainly focused on commonly used AI tools, such as ChatGPT® and Grammarly®, without considering other AI applications relevant to pharmacy education, such as AI-driven

drug discovery platforms or clinical decision support systems. This focus may limit the scope of the findings in terms of students' knowledge and perceptions of the broader AI landscape.

Moreover, the online survey format could have introduced selection bias, as students more comfortable with technology or those with a greater interest in AI may have been more inclined to participate, possibly skewing the results. Lastly, the cross-sectional nature of this study captures students' knowledge and perceptions at a single point in time, which may not reflect how these views evolve as AI technologies advance. Longitudinal studies could provide more insight into how students' knowledge and perceptions change over time.

Conclusion

This study provides insights into pharmacy students' knowledge and perceptions of chat-based AI tools in pharmacy education. The findings suggest that students generally have a good understanding of these tools and view them as integral components of modern educational practices. However, there remain misconceptions among some students about the roles and capabilities of AI technologies, highlighting a need for targeted educational interventions.

As AI tools become more prevalent in healthcare and education, it is essential for pharmacy curricula to keep pace, ensuring that students are well-prepared for their future roles. Future studies should explore how AI applications, beyond chat-based tools, could enhance pharmacy education and practice.

Abbreviations

AI	Artificial Intelligence
CHATGPT	Chat Generative Pre-Trained Transformer
ABUAD	Afe Babalola University Ado-Ekiti
ABUADHREC	Afe Babalola University Health Research and Ethics Committee

Supplementary Information

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Supplementary Material 1

Author contributions

All authors contributed to the study conception and design. Material preparation, data collection were performed by EO, OE and PC. Data analysis was performed by EO, CO, BE and EE. The first draft of the manuscript was written by EO, PC, AO, MK, CO, MO, SE. OA and IE reviewed the manuscript. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability

Data material used in this study will be made available by the corresponding authors on request.

Declarations

Ethical approval

This study obtained ethical approval from the Afe Babalola University Health Research and Ethics Committee (ABUADHREC) on 2/3/2024 before commencement, and the approval number was ABUADHREC/19/05/2024/24. Informed consent was obtained from all subjects and/or their legal guardian(s). This study was conducted while assuring the participants about the confidentiality of every piece of information provided to the participants. All methods were carried out following relevant guidelines and regulations.

Consent for publication

Not Applicable.

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

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